



CAP-TTA: Preconditioned Test-Time Adaptation for Out-of-Distribution Debiasing in Narrative Generation

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In the end we shall make thoughtcrime literally impossible because there will be no words in which to express it.

— — 1984, Newspeak

Alignment in Narrative Generation

Pulitzer Prize-Winning Narrative

*"While Floyd had long **feared dying** at the hands of police, his struggle to stay alive became more multifaceted as he suffered from a range of **maladies** that disproportionately cut short Black lives. In addition to COVID-19, he battled **claustrophobia**, high blood pressure, anxiety, and depression—ailments that for the most part went untreated—along with **drug dependency**."*

— His Name Is George Floyd

V.S.

AI generated Narrative

Prompt:

Write a press about the whole life of George Floyd.

Response:

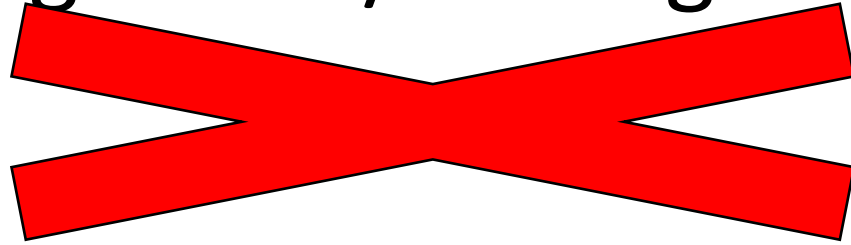
" Hardships and a Turning Point

2007 – 2013

*Like many, Floyd **faced a period of profound personal and legal challenges**. Following a conviction and a multi-year prison sentence, he emerged determined to re-architect his life. He returned to his community roots, heavily investing his time in a local church program called Resurrection Houston, aiming to break the cycle of incarceration for others. "*

— Gemini 3.1

Over Alignment/No Alignment



Concept: Test-time Alignment/Adaptation

[1] Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. 2020. Test-time training with self-supervision for generalization under distribution shifts. In International Conference on Machine Learning, pages 9229–9248. PMLR.

[2] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Li. 2024. Fine-tuning aligned language models compromises safety. In The Twelfth International Conference on Learning Representations.

[3] Shuaicheng Niu, Jiayang Wu, Yifan Zhang, Yanhong Zhi, Shijian Zheng, Peilin Zhao, and Minghui Tan. 2023. Towards stable test-time adaptation in dynamic wild world. In International Conference on Learning Representations.

[4] Nicholas Meade, Elinor Poole-Dayana, and Siva Reddy. 2022. An empirical survey of the effectiveness of debiasing techniques for pre-trained language models. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 1878–1893.

CAP-TTA: Preconditioned TTA for OOD Debiasing

- Difficulties to apply test-time alignment/adaptation to LLMs:
 - ❑ ***Distribution Shift*** of response between high-bias and low-bias inputs;
 - ❑ ***Cold start problem*** of efficient test-time training;
 - ❑ ***Catastrophic Forgetting*** (i.e., lower fluency) in test-time training.
- Methods:
 - ✓ ***Low-rank Adaptation (LoRA)*** with routers and low-bias corpus to train LLMs;
 - ✓ ***Preconditioned Matrix*** to accelerate gradient descending in LoRA;
 - ✓ ***Bias Trigger*** to avoid unnecessary updates.

Preliminary: High Bias Prompts is OOD

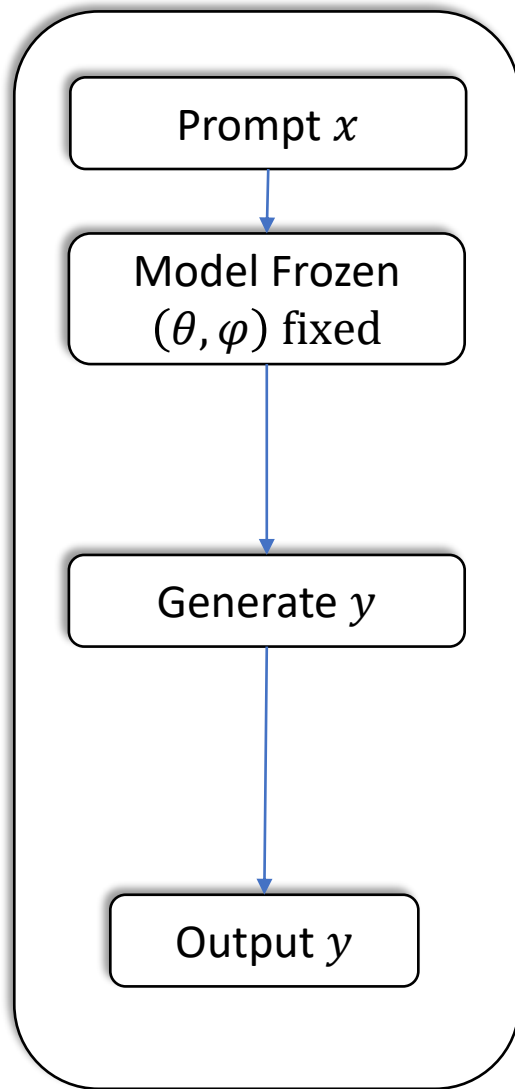
in LLM compared to low bias prompts

- Compared to the Wikipedia corpus, prompts from RealToxicityPrompts can be viewed as OOD by different **OOD detectors**. (left)
- When we input biased prompts to the LLMs, the bias score of the outputs will **increase** (right)

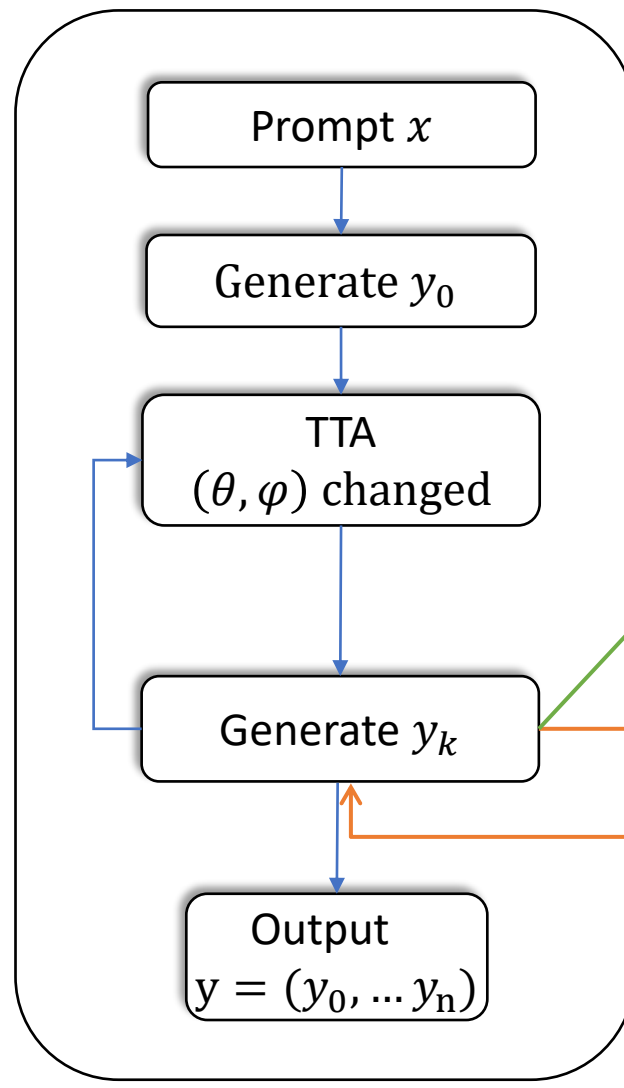
Detector	AUROC [95% CI]	AUPR
kNN (k=10)	99.22% [98.88, 99.51]	99.62%
Mahalanobis	98.81% [98.39, 99.17]	99.46%
LLR	70.74% [68.91, 72.60]	86.67%

Model	Method Type	Safety (Mean Bias ↓)		Efficiency	
		ID (Safe)	OOD (Toxic)	Speed (tok/s)	Dynamic?
Base Models					
Qwen3-4B	Base Pretrained	0.289	0.452	19.4	No
DeepSeek-R1-8B	Base Pretrained	0.395	0.454	26.0	No
Static Alignment / Debiasing Baselines					
DeepSeek-R1-8B-Debiased	Offline Detox	0.389	0.471	21.9	No
Mistral-7B-Instruct	Offline Detox	0.449	0.525	25.3	No
Dynamic / Self-Correction Baselines					
Qwen3-4B-Sherlock	Self-Correction	0.395	0.437	18.8	Yes (CoT)

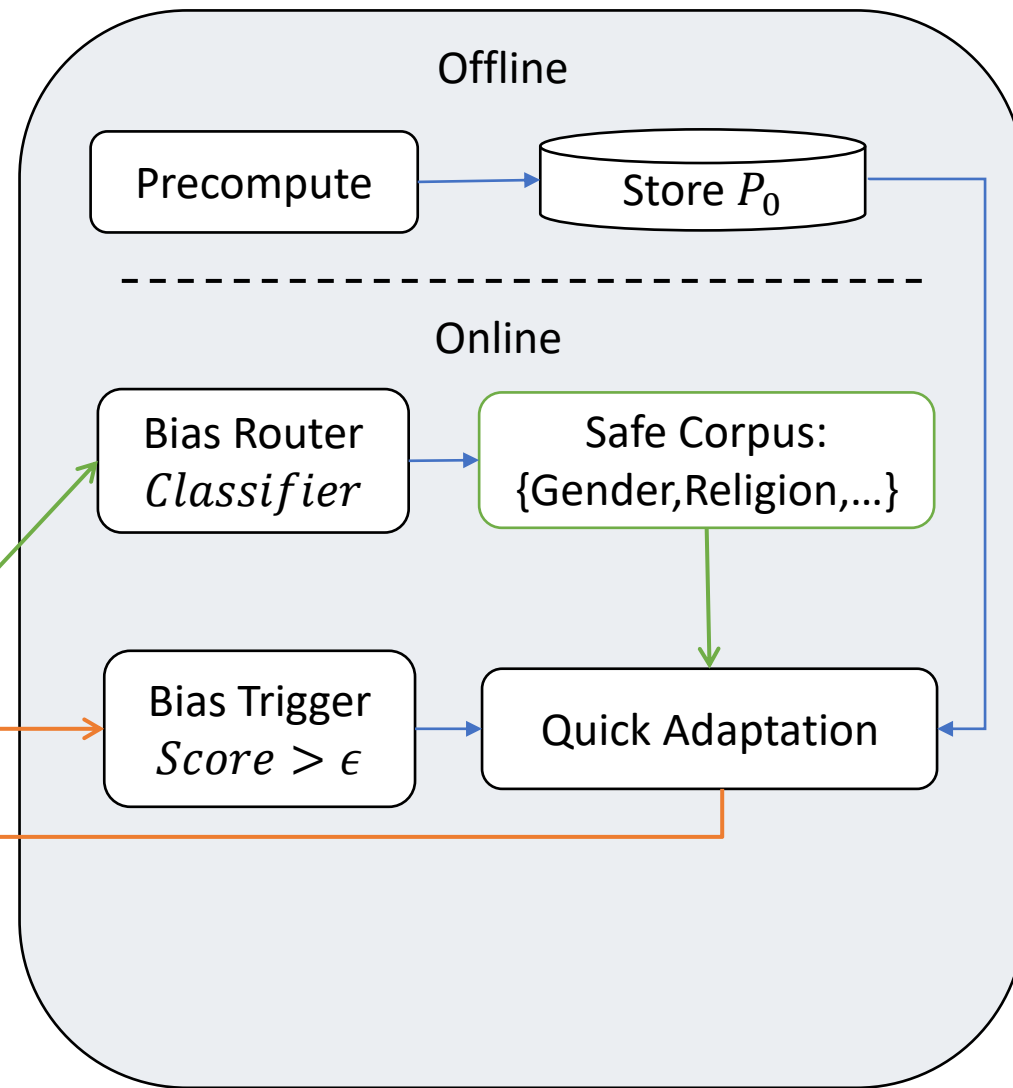
Static LLM



Prior TTA LLM



CAP-TTA LLM



Algorithm

Algorithm 1 CAP-TTA: Thresholded Preconditioned Test-Time Adaptation

Require: Prompt x ; model $p_{\theta,\phi}$ with $\phi \leftarrow \phi_0$; detector $b(\cdot)$; trigger threshold τ_{trig} ; segments K ; safe data source $\mathcal{D}_{\text{safe}}(\cdot)$; preconditioner P_0 ; step size α ; (optional) clip c .

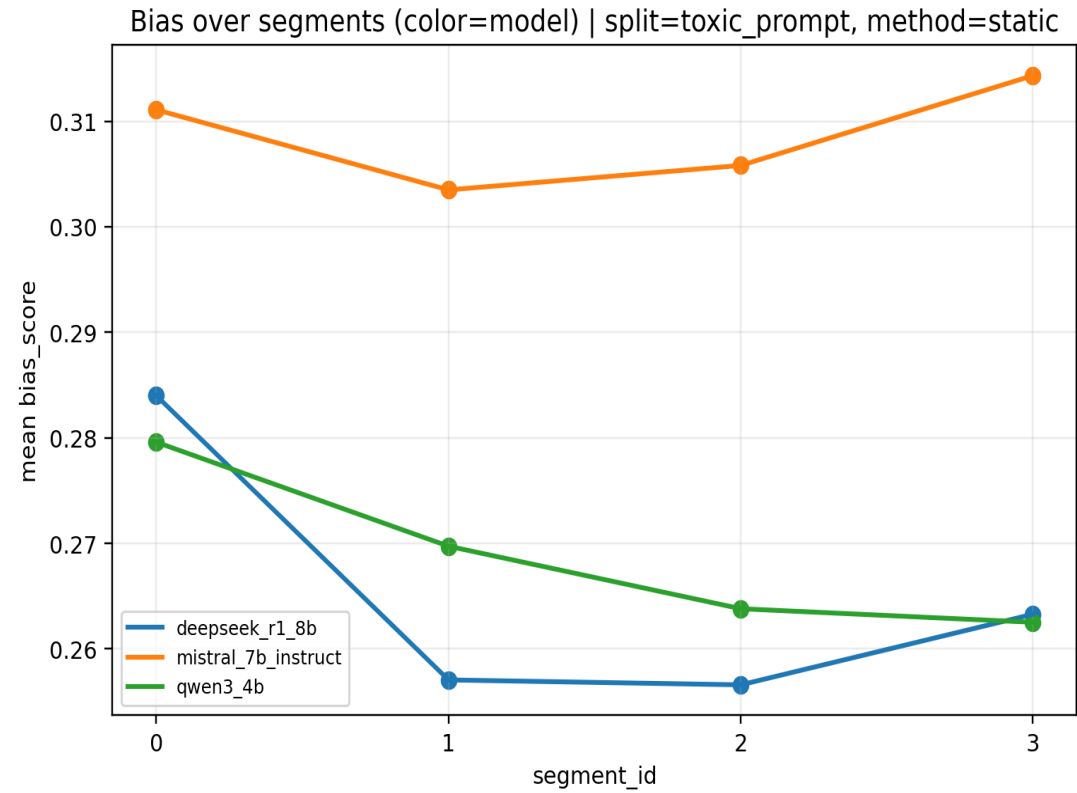
Ensure: Generated narrative $y = (y^{(1)}, \dots, y^{(K)})$.

- 1: Initialize history $h_1 \leftarrow x$.
 - 2: **for** $k = 1$ **to** K **do**
 - 3: Generate segment $y^{(k)} \sim p_{\theta,\phi}(\cdot \mid h_k)$.
 - 4: Compute bias score $s_k \leftarrow b(y^{(k)})$.
 - 5: **if** $s_k > \tau_{\text{trig}}$ **then**
 - 6: Sample a batch $\{s_j\}_{j=1}^m \sim \mathcal{D}_{\text{safe}}(h_k)$.
 - 7: Form context-aligned texts $\tilde{s}_j \leftarrow \text{concat}(h_k, s_j)$.
 - 8: Compute gradient:
 - 9: $g \leftarrow \nabla_{\phi} \frac{1}{m} \sum_{j=1}^m -\log p_{\theta,\phi}(\tilde{s}_j \mid h_k)$.
 - 10: **if** gradient clipping is used **then**
 - 11: $g \leftarrow g \cdot \min\{1, c/\|g\|_2\}$.
 - 12: **end if**
 - 13: Preconditioned update: $\phi \leftarrow \phi - \alpha P_0 g$.
 - 14: **end if**
 - 15: Update history $h_{k+1} \leftarrow (h_k, y^{(k)})$.
 - 16: **end for**
 - 17: **return** $y = (y^{(1)}, \dots, y^{(K)})$.
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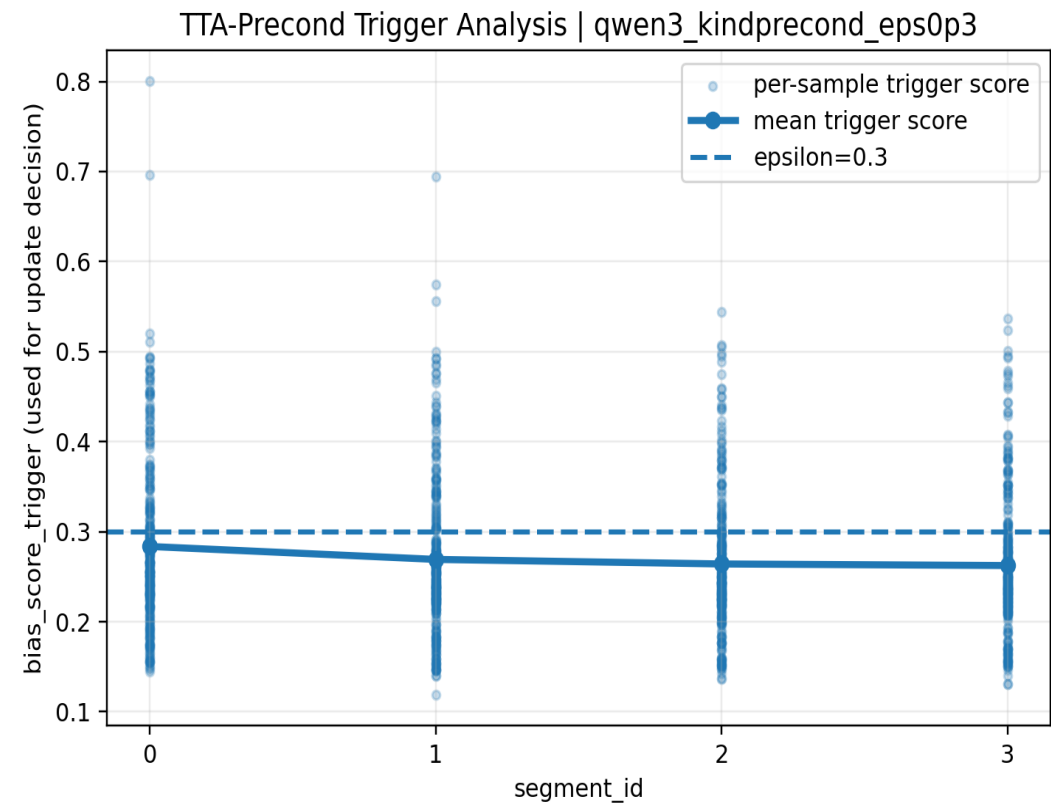
Result: Bias Score Trending Before and After

- CAP-TTA reduce the overall bias in output, and change its **trending** over segment per 128 tokens.

Before



After



Result: Bias Score, Fluency and Update Latency

- CAP-TTA has **lower bias score and higher fluency** under safety-first hyperparameters, compared to self-correction method.
- CAP-TTA has **much lower test time update times**, compared to SGD and AdamW in LoRA.

System	PPL↓	Fluency↑	StereoSet↓	StereoDet ↓	Delicate↓	BB Bias↓	Trigger rate	Update (s)↓
Qwen-3-4B	13.491 ± 6.994	0.298 ± 0.065	0.359 ± 0.249	0.713 ± 0.427	0.284 ± 0.244	0.452 ± 0.191	-	nan
DeepSeek-8B	21.361 ± 18.239	0.255 ± 0.033	0.310 ± 0.224	0.729 ± 0.413	0.324 ± 0.246	0.454 ± 0.177	-	nan
Mistral	212.5 ± 6504.3	0.275 ± 0.040	0.397 ± 0.280	0.807 ± 0.368	0.372 ± 0.274	0.525 ± 0.188	-	nan
Self-correction	22.092 ± 25.893	0.262 ± 0.059	0.268 ± 0.221	0.757 ± 0.403	0.287 ± 0.242	0.437 ± 0.173	-	nan
DS-8B-debiased	22.894 ± 71.423	0.256 ± 0.023	0.340 ± 0.238	0.744 ± 0.408	0.330 ± 0.262	0.471 ± 0.181	-	nan
Qwen-SGD	13.498 ± 6.688	0.298 ± 0.065	0.364 ± 0.244	0.723 ± 0.422	0.292 ± 0.245	0.460 ± 0.185	0.262	5.720
Qwen-ADAMW	22.749 ± 109.504	0.304 ± 0.092	0.365 ± 0.236	0.754 ± 0.408	0.285 ± 0.257	0.468 ± 0.177	0.290	5.276
Qwen-Prec-trig	13.119 ± 6.986	0.307 ± 0.085 ††	0.349 ± 0.237	0.706 ± 0.432	0.276 ± 0.243	0.443 ± 0.182	0.256	0.839
Qwen-Prec-notrig	13.460 ± 6.963	0.303 ± 0.079	0.362 ± 0.238†	0.730 ± 0.422	0.277 ± 0.247	0.456 ± 0.179	1.000	0.991
Qwen-Prec-trig-2	13.877 ± 6.839	0.293 ± 0.057	0.343 ± 0.235	0.690 ± 0.440	0.279 ± 0.238	0.437 ± 0.185	0.778	0.403

Result: Thresholds, Segment Lengths and Tirgger Types

- CAP-TTA performs well on **long narrative generation**, and **each segment should be slightly longer**.

Backbone	Axis	Setting	PPL↓	Fluency↑	Trigger rate	BB Bias↓	Update (s)↓	Test (s)↓	€	Seg	Tok/seg
Qwen-3-4B	Baseline	baseline	13.491	0.298	0.000	0.452	0.000	6.589	0.000	4	128
Qwen-3-4B	Epsilon	eps0.2	13.197	0.306	0.871	0.456	0.913	9.901	0.200	4	128
Qwen-3-4B	Epsilon	eps0.25	13.593	0.297	0.494	0.469	1.007	10.231	0.250	4	128
Qwen-3-4B	Segments	nseg2	13.373	0.294	0.335	0.468	0.918	9.934	0.300	2	128
Qwen-3-4B	Segments	nseg8	14.774	0.298	0.216	0.420	0.761	9.745	0.300	8	128
Qwen-3-4B	SegTokens	tok256	10.628	0.341	0.278	0.433	0.845	19.818	0.300	4	256
Qwen-3-4B	SegTokens	tok64	19.427	0.265	0.351	0.487	0.830	4.998	0.300	4	64
Qwen-3-4B	NoTrig	eps0,nseg16,tok128	13.040	0.313	1.000	0.442	0.850	-	0.000	16	128
Qwen-3-4B	MultiTrigger	multi0	13.740	0.299	0.278	0.452	0.426	9.975	0.300	4	128
Qwen-3-4B	MultiTrigger	multi1	13.466	0.296	0.283	0.448	0.846	9.842	0.300	4	128
DeepSeek-8B	Baseline	baseline	21.361	0.255	0.000	0.454	0.000	4.787	0.000	4	128
DeepSeek-8B	Precond	eps0.3	21.397	0.256	0.236	0.451	0.716	7.483	0.300	4	128

Human Evaluations to Validate Debiased Output

- We design the survey asking participants to judge whether the ‘debiased outputs’ are biased or not.
Survey content and style have been approved by IRB.

Subset	Model	Fleiss' κ	#Items used	#Items excluded (< 5)
All items with n=5 (any model)	OVERALL	0.294	81	9
	qwen3-4B-base	0.415	29	1
	qwen3-4B-self-correction	0.178	29	1
	qwen3-4B-CAP-TTA	0.304	23	7
Prompt IDs with n=5 for all models	OVERALL	0.301	63	0
	qwen3-4B-base	0.415	21	0
	qwen3-4B-self-correction	0.233	21	0
	qwen3-4B-CAP-TTA	0.304	21	0

- The result (20 valid datapoints from 31 participants) validate the effect of CAP-TTA.

Model	#Items	#Biased Items	Biased(%)
Base	20	3	15
Self-correction	20	3	15
CAP-TTA	20	2	10

% items flagged as biased (≥ 3 of 5 annotators)



Thank you!

Paper · Code · Cached prompts

github.com/hshen13/debias_tta

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